

BUS 123 - Solving Business Problems with Technology

MATH-M09-L02

BUS 123 — MATH-M09-L02 — PMT Function & Annuities

PRE-READING

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BUS123 · MATH-M09 · L02 PRE-READING

PMT FUNCTION & ANNUITIES

Course: Solving Business Problems with Technology · BUS123 **Track:** MATH · Module 09 · Lesson 02

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CONNECT TO PRIOR KNOWLEDGE

In M09-L01 we used `=FV()` to grow a single lump sum forward in time. In every problem, the `pmt` argument was set to zero — because there were no recurring payments, only a one-time deposit.

Today that changes. In the real world, most financial decisions involve **regular payments**: you pay your mortgage every month, contribute to your retirement account every paycheck, and collect a pension every quarter. These payment streams are called **annuities**, and they are the most common financial structure in everyday life.

The good news: the same three Excel functions — `=PV()`, `=FV()`, and the new `=PMT()` — handle everything. The only change is that `pmt` is no longer zero. The sign rules and the period-matching rule from L01 carry forward unchanged.

CORE CONCEPTS

Part A — What Is an Annuity?

An **annuity** is a series of equal payments made at equal time intervals. Every mortgage payment, car loan installment, monthly retirement contribution, and lease payment is an annuity. Annuities are the most common financial structure in personal and business finance.

Two types based on payment timing:

Type	When payments occur	Excel <code>type</code> argument	Common uses
Ordinary Annuity	End of each period	0 (default — omit)	Loans, mortgages, most savings plans
Annuity Due	Beginning of each period	1	Lease payments, insurance premiums

Annuity due always produces a slightly higher FV or lower required PMT than ordinary annuity, because each payment earns one extra period of interest. Unless a problem specifically states "beginning of period" or uses the word "due," assume ordinary annuity (`type = 0`).

Two types based on what you're solving for:

- - **Loan amortization:** You know the loan amount (PV) — you want to know the required periodic payment (PMT).
- - **Savings goal:** You know the target balance (FV) — you want to know the required periodic contribution (PMT).

Both use `=PMT()`. The difference is which argument holds the target value.

Part B — The `=PMT()` Function

`=PMT()` calculates the periodic payment required to pay off a loan or reach a savings goal over a fixed number of periods at a fixed interest rate.

Syntax:

`=PMT(rate, nper, pv, [fv], [type])`

Argument	Meaning	For loan problems	For savings goal problems
rate	Rate per period	Annual rate ÷ 12 (monthly)	Annual rate ÷ 12 (monthly)
nper	Total payment periods	Years × 12 (monthly)	Years × 12 (monthly)
pv	Present value (loan amount)	Loan amount — enter positive	Enter 0
fv	Future value (savings target)	Enter 0 (loan paid off)	Target balance — enter positive
Result	Required payment	Returns negative	Returns negative

The result is always **negative** — it represents money leaving your pocket each period (an outflow). When communicating the monthly payment to a client, use the absolute value: `=ABS(PMT(...))`.

Worked Example — Meridian Advisory Group Mortgage: A client purchases a home for \$320,000 at 6.5% annual interest, 30-year term, monthly payments.

- - `rate = 6.5% ÷ 12 = 0.5417% per month`
- - `nper = 30 × 12 = 360 months`
- - `=PMT(6.5%/12, 360, 320000, 0) → -$2,022.62/month`

The client pays **\$2,022.62 per month**. Over 30 years:

- - `Total paid = ABS(-$2,022.617675...) × 360 = $728,142.36`
- - `Original PV = $320,000`
- - `Total interest paid = $728,142.36 - $320,000 = $408,142.36`

The client will pay more than \$400,000 in interest over the life of the loan — more than the original principal. Keep the unrounded PMT in the formula cell and round only the display; multiplying the displayed \$2,022.62 by 360 creates a small rounding difference.

Build a Labeled PMT Model in Excel

Use a compact input-and-output model instead of typing numbers directly into formulas:

Cell	Label	Enter	Format
B4	Loan amount	320000	Currency, 2 decimals
B5	Annual rate	6.5%	Percentage, 2 decimals
B6	Years	30	Number, 0 decimals
B7	Payments per year	12	Number, 0 decimals
B8	Monthly payment	formula	Currency, 2 decimals
B9	Total paid	formula	Currency, 2 decimals
B10	Total interest	formula	Currency, 2 decimals

1. Select **B8**, type `=PMT(B5/B7,B6*B7,B4,0)`, and press **Enter**. Expected result: **-\$2,022.62**.
2. Select **B9**, type `=ABS(B8)*B6*B7`, and press **Enter**. Expected result: **\$728,142.36**.
3. Select **B10**, type `=B9-B4`, and press **Enter**. Expected result: **\$408,142.36**.
4. On Windows Excel, use **Home > Number** and apply Currency with 2 decimal places to B4 and B8:B10. Mac Excel may position these commands differently.
5. Reasonableness checks: PMT should be negative because it is a payment; total paid must exceed the loan amount; total interest must be positive.
6. Recalculation test: change **B5 from 6.5% to 7.0%**. The payment should become more negative and total interest should increase automatically. Then restore B5 to 6.5%.

Excel for Windows - mortgage model

Formula bar: `=PMT(B5/B7,B6*B7,B4,0)`

Loan amount	\$320,000
Annual rate	6.50%
Years	30
Payments/year	12
Monthly payment	-\$2,022.62
Total interest	\$408,142.36

Output formulas

B8: `=PMT(B5/B7,B6*B7,B4,0)`

B9: `=ABS(B8)*B6*B7`

B10: `=B9-B4`

Use Currency with 2 decimals.

Negative PMT means cash paid out.

Cash-flow signs and automatic recalculation

Money received = positive

Loan proceeds: PV = +320000

Settlement checks: PMT = +1500

Excel balances with the opposite sign.

Input-change test

Change 6.5% to 7.0%.

PMT becomes more negative.

Total interest increases.

The layout mirrors the starter workbook: use labeled yellow input cells, enter formulas in the answer/output cells, and check the feedback in **Live You Try It**. The **FormulaReferenceCard** is a syntax reminder, not a completed solution. Build the graded **Class Challenge** formulas yourself.

⚠ The Most Common PMT Error: Wrong Period Match

`=PMT(6.5%, 30, 320000)` returns $-\$27,006$ — a large *annual* payment that looks plausible. But the monthly payment is \$2,023. The formulas are not wrong; the *inputs* are mismatched. Always ask: what is the payment frequency? Rate and nper must match that frequency.

Part C — Solving for Required Savings Contribution

`=PMT()` also works in reverse: given a savings goal (FV), it tells you how much to contribute each period.

Worked Example — Retirement Goal: A Meridian client wants \$500,000 in 25 years. Their account earns 7% annual, compounded monthly. How much must they contribute each month?

`=PMT(7%/12, 25*12, 0, 500000)` → $-\$576.57/\text{month}$

They must contribute **\$576.57 per month**. Over 25 years:

- Total contributed = $\$576.57 \times 300 = \$172,971$
- Account balance at goal = $\$500,000$
- Growth from compounding = $\$327,029$

They put in \$172,971 and compounding nearly doubled it to \$500,000. The gap — \$327,029 — came entirely from interest earning interest over 25 years.

Part D — =FV() with Regular Contributions

When a client makes regular monthly contributions, `=FV()` projects the ending balance. The `pmt` argument is no longer zero.

Syntax (same as before, pmt now filled in):

`=FV(rate, nper, pmt, [pv], [type])`

For a savings account with monthly contributions and no starting balance:

- - `rate` = annual rate ÷ 12
- - `nper` = years × 12
- - `pmt` = monthly contribution — enter as **negative** (money leaving your pocket)
- - `pv` = 0 (no starting lump sum)

Worked Example: A Meridian client contributes \$500/month to a retirement account earning 7% annual for 30 years.

`=FV(7%/12, 30*12, -500, 0)` → \$609,985.50

They contributed $\$500 \times 360 = \$180,000$ total. Compounding grew it to **\$609,985.50** — the extra **\$429,985.50** is entirely from compound interest on prior contributions.

Starting with both a lump sum and regular contributions:

If the client already has \$10,000 saved and also contributes \$500/month:

`=FV(7%/12, 360, -500, -10000)` → \$691,150.47

Both `pmt` and `pv` are negative because both are outflows — money leaving the client's pocket.

Part E — =PV() with Regular Payments

`=PV()` with a `pmt` value answers: *what is a stream of equal future payments worth today?*

Worked Example: A Meridian client will receive \$1,500/month for 10 years from a structured settlement. Discount rate: 5% annual. What is that stream worth in today's dollars?

`=PV(5%/12, 10*12, 1500, 0)` → -\$141,422.03

The stream pays out $\$1,500 \times 120 = \$180,000$ total in nominal dollars. In today's dollars at 5%, it is worth only **\$141,422.03**. The difference — **\$38,577.97** — represents the time value eroded by 10 years of waiting and discounting.

The result is negative because it represents the **cost to acquire this stream today** (outflow). If a buyer offers \$130,000 for the settlement, the client should hold — it is worth more. If the offer is \$155,000, the client should accept — the offer exceeds the present value.

Part F — The Sign Convention Across All Three Functions

The same cash-flow rule applies in every TVM function:

- - **Money leaving your pocket** (investing, paying a loan, making a deposit) → **negative**
- - **Money coming to you** (receiving a loan, collecting a payment, withdrawing savings) → **positive**

Scenario	pvt sign	pmt sign	fv sign
Taking out a loan	+ (received)	- (paying back)	0
Regular savings contributions	0 or - (initial deposit)	- (monthly deposit)	+ (balance received)
Valuing received annuity payments	0	+ (payments received)	0

Getting any one of these signs wrong does not cause a formula error — it produces a plausible-looking answer with the wrong sign or direction. Always verify: does the result make economic sense given what you entered?

FORMULA REFERENCE TABLE

Formula	Use
<code>=PMT(rate/12, years×12, pv, 0)</code>	Monthly loan payment (pv = loan amount, positive)
<code>=PMT(rate/12, years×12, 0, fv)</code>	Required monthly savings contribution (fv = target)
<code>Total interest = ABS(PMT) × nper - PV</code>	Total interest cost over loan life
<code>=FV(rate/12, years×12, -pmt, 0)</code>	Future savings balance from regular contributions
<code>=FV(rate/12, years×12, -pmt, -pv)</code>	Future savings with starting balance + contributions
<code>=PV(rate/12, years×12, pmt, 0)</code>	Present value of a payment stream (pmt positive = received)

CHECK YOUR UNDERSTANDING

Answer these questions before class. Show your work on questions 2–6.

1. In your own words, what is an annuity? Give two examples from everyday life — one involving money flowing out (a loan) and one involving money flowing in (a payment received).
2. A Meridian client takes out a \$200,000 mortgage at 5.5% annual interest for 15 years, with monthly payments. (a) Write the `=PMT()` formula. (b) What is the monthly payment? (c) What is the total interest paid over the life of the loan?
3. A client wants to accumulate \$300,000 in 20 years. Their account earns 6% annual, compounded monthly. How much must they contribute each month? Write the `=PMT()` formula and identify which argument holds the savings goal.
4. A Meridian client contributes \$400/month to an account earning 6% annual for 25 years, starting with no existing balance. Use `=FV()` to project the ending balance. What is the total amount contributed, and how much of the balance came from compounding?
5. A client will receive \$2,000/month for 12 years from a structured settlement. The current discount rate is 6% annual. Use `=PV()` to find the present value of this stream. Is a lump-sum offer of \$170,000 a good deal for the client — or should they collect the payments?
6. True or False: In `=PMT()`, the result is always negative. Explain what the negative sign means economically.
7. A client is weighing two options over 10 years at 5% annual (monthly compounding): (a) invest \$40,000 today as a lump sum, or (b) contribute \$350/month with no starting balance. Which produces a higher account balance at year 10? Show both calculations.

ANSWER KEY

1. An annuity is a series of equal payments made at equal time intervals. Examples: outflow — monthly car loan or mortgage payments (money leaves your pocket each month). Inflow — monthly pension payments or a structured settlement (money arrives each month).

2. (a) $\text{=PMT}(5.5\%/12, 15*12, 200000, 0)$. (b) Monthly payment = **\$1,634.17**. (c) Total paid = $\$1,634.17 \times 180 = \$294,150.60$. Total interest = $\$294,150.60 - \$200,000 = \text{\$94,150.60}$.

3. $\text{=PMT}(6\%/12, 20*12, 0, 300000)$ → **-\$671.93/month**. The savings goal lives in the `fv` argument (the fourth argument). `pv` is 0 because the client is not starting with a lump sum.

4. $\text{=FV}(6\%/12, 300, -400, 0)$ → **\$401,806.46**. Total contributed = $\$400 \times 300 = \$120,000$. Growth from compounding = $\$401,806 - \$120,000 = \text{\$281,806}$. Compounding produced more than twice the total contributions.

5. $\text{=PV}(6\%/12, 144, 2000, 0)$ → **-\$200,904.37**. The stream is worth \$200,904 in today's dollars. A lump-sum offer of \$170,000 is **below** present value — the client should decline and collect the payments. If offered more than \$200,904, the client should consider accepting.

6. **True**. The $\text{=PMT}()$ result is always negative because it represents money leaving the borrower's or saver's pocket each period — an outflow. For a loan, you are paying the lender. For a savings goal, you are making a deposit. In both cases the cash flows in the negative direction from your perspective. Report the payment to clients as $\text{ABS}(\text{PMT}())$.

7. (a) Lump sum: $\text{=FV}(5\%/12, 120, 0, -40000)$ = **\$65,926.88**. (b) Monthly contributions: $\text{=FV}(5\%/12, 120, -350, 0)$ = **\$54,428.77**. The lump sum wins by **\$11,498** because the full \$40,000 compounds for all 10 years, while the \$350/month contributions average only about 5 years of compounding. Starting early with a larger amount generally beats smaller regular contributions over the same horizon at the same rate.

KEY VOCABULARY

Term	Definition
Annuity	A series of equal payments made at equal time intervals.
Ordinary Annuity	Payments made at the end of each period. The Excel default (type = 0). Used for most loans and savings plans.
Annuity Due	Payments made at the beginning of each period (type = 1). Used for leases and some insurance products.
=PMT()	Excel function that calculates the required periodic payment given rate, nper, and either PV (loan) or FV (savings goal).
Loan Amortization	The process of paying off a loan through regular equal payments, each of which covers some interest and some principal.
Total Interest	The total cost of borrowing: $\text{ABS(PMT)} \times \text{nper} - \text{PV}$.
Savings Contribution	A periodic deposit toward a future savings goal. In =FV() and =PMT(), entered as a negative number (outflow).
pmt (argument)	The recurring payment per period in Excel TVM functions. Negative for money leaving your pocket; positive for money received.
Sign Convention	In Excel TVM functions, outflows are negative and inflows are positive. Applies to pv, pmt, and fv.
Structured Settlement	A financial arrangement that pays a fixed amount at regular intervals — a real-world annuity commonly valued using =PV().
